

Sharecoin: A Bitcoin-Derived Blockchain Providing Verifiable Randomness via GPU-Based Proof-of-Work

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Abstract

Sharecoin is a Bitcoin-Core-derived blockchain that repurposes GPU-based Proof-of-Work (PoW) as a decentralized randomness beacon. By replacing Bitcoin's SHA-256d PoW with the memory-hard KawPow algorithm, Sharecoin exposes the per-block `mix_hash` as an unpredictable, economically expensive, and publicly verifiable source of entropy. The protocol combines the `mix_hash` values of consecutive blocks into a randomness value accessible through RPC. The value cannot be known before those blocks are mined, and the amount a miner can bias it is bounded by that miner's share of the hashrate. Sharecoin preserves full Bitcoin Core compatibility (including P2P networking, validation logic, wallet infrastructure, and RPC semantics) while introducing a novel public-good utility: publicly verifiable randomness generation. This whitepaper formally describes the protocol, threat model, implementation details, and applications.

1 Introduction

Randomness is a foundational primitive in distributed systems, cryptography, gaming, lotteries, governance, and decentralized applications. Existing blockchain ecosystems lack a native, trustless randomness source. Smart-contract platforms typically rely on centralized oracles, commit-reveal schemes vulnerable to last-revealer bias, VRF systems requiring trusted key-holders, or external randomness services.

Sharecoin introduces a new approach: mining produces randomness as a public good. By leveraging the inherent unpredictability of GPU-based PoW, Sharecoin transforms the mining process into a decentralized randomness beacon without introducing trusted parties or additional cryptographic assumptions.

Sharecoin is a full Bitcoin Core fork with minimal, auditable modifications. The protocol retains Bitcoin's security model, consensus rules, and network architecture while replacing SHA-256d with KawPow and exposing the resulting entropy.

2 Background and Motivation

2.1 Bitcoin's PoW and Determinism

Bitcoin's SHA-256d PoW is deterministic once a block is mined. The block hash is unpredictable prior to discovery but is not suitable as a randomness beacon because ASIC mining is highly optimized and predictable, miners can cheaply grind nonces, and block hash manipulation is economically feasible.

2.2 GPU-Based PoW as an Entropy Source

Memory-hard PoW algorithms such as KawPow incorporate large memory reads, data-dependent access patterns, GPU architecture variability, and unpredictable mix-state evolution. The `mix_hash` produced during KawPow computation is expensive to influence, unpredictable prior to computation, and infeasible to bias without re-mining the entire block. This makes it a suitable entropy source.

2.3 Design Goals

Sharecoin aims to:

- Provide trustless, verifiable randomness secured by PoW.
- Preserve Bitcoin Core compatibility and security.
- Maintain ASIC resistance and decentralization.
- Offer a transparent, reproducible codebase with minimal diffs.

3 System Architecture

3.1 Consensus Layer

Sharecoin inherits Bitcoin's consensus rules with the following modifications:

- PoW algorithm: KawPow replaces SHA-256d.
- Block header includes `mix_hash`.
- Difficulty adjustment uses LWMA (Linearly Weighted Moving Average), recalculating the target on every block from a weighted average of recent solvetimes, rather than Bitcoin's original fixed 2016-block retargeting window. This reacts within roughly a window's worth of blocks to hashrate changes, relevant for a network with a smaller and more variable miner base than Bitcoin's.

3.2 Mining Process

Mining proceeds as follows:

1. Miner constructs a candidate block.
2. KawPow is executed using the block header and DAG.
3. The algorithm produces:
 - `mix_hash`: internal state hash
 - `result`: final PoW output
4. If `result < target`, the block is valid.
5. `mix_hash` is committed into the block header.

3.3 Randomness Beacon

Sharecoin exposes randomness via:

- Per-block entropy: `mix_hash`
- Aggregated entropy: RPC `getrandombeacon`

The beacon aggregates the last N confirmed blocks:

$$R = H(\text{mix_hash}_i \parallel \text{mix_hash}_{i-1} \parallel \cdots \parallel \text{mix_hash}_{i-N+1})$$

The value cannot be known before the last block of the window is mined. The miner who finds that block, however, sees the value first and can discard the block and try again. Section 4.2 bounds what that is worth. The window size does not change it.

4 Security Analysis

4.1 Threat Model

Adversaries may attempt:

- Biasing randomness by selective block publication.
- Grinding attacks by re-mining blocks.
- 51% attacks to control beacon output.

4.2 Bias Resistance

The miner who finds the block that completes the window sees the finished value before deciding whether to publish it. If they dislike it they can discard the block, which forfeits its reward, and the race for that height starts again. This holds for every window size, including a window of one block.

Let a miner control a fraction a of the hashrate, and let q be the probability that the outcome they want occurs by chance. The probability that they get it is

$$P = \frac{q}{1 - a(1 - q)},$$

so they can multiply their odds by at most $1/(1 - a)$. For a fair coin flip ($q = 1/2$), $a = 10\%$ gives 52.6%, $a = 33\%$ gives 59.9% and $a = 49\%$ gives 66.2%. A simulation of the real construction (SHA-256 over the concatenated `mix_hash` values) gives the same result for window sizes 1, 2, 8 and 100.

The bound assumes that (i) no miner holds a majority of the hashrate, (ii) the inputs being drawn from were committed before the completing block was found, and (iii) nobody holds several of the last blocks of the window privately, which takes luck of about a^k for k blocks. The cost of withholding is the block reward given up. SHC has no market price, so we make no claim about that cost in money. The live share of the largest miner is published at sharecoin.cc/beacon, and the simulation is in `docs/analysis/beacon_bias_sim.py`.

A delay after the value is known helps against withholding: the miner then has to decide before they can learn the result, and if another miner finds a block at that height in the meantime theirs is orphaned. In simulation, with 120 second blocks and a miner holding 33% of the hashrate, a delay of 5 minutes leaves about 20% of the bias and 10 minutes leaves about 5%.

4.3 Grinding Resistance

Grinding requires finding further valid blocks, and every retry costs a full block of work. The number of retries an attacker gets is therefore limited by their share of the hashrate, which is the same bound as above.

4.4 Majority Miners

A miner with a majority of the hashrate can rewrite the window and retry repeatedly, so the bound above does not apply. This is a real limitation while the network's hashrate is small and concentrated. At the time of this revision the largest miner held more than half of the blocks in most 100-block windows over a four day period.

5 Implementation Details

5.1 Codebase Structure

Sharecoin maintains a minimal diff from Bitcoin Core:

- PoW replaced with KawPow implementation from Ravencoin.
- Block header extended to include `mix.hash`.
- RPC additions: `getrandombeacon`.
- Build system preserved (CMake, vcpkg, autotools).

Patch scripts allow deterministic reproduction of diffs.

5.2 Wallet and Node Software

Sharecoin provides:

- `sharecoind` (daemon)
- `sharecoin-cli` (RPC client)
- `sharecoin-qt` (GUI wallet)
- Windows binaries
- Linux build instructions

5.3 Mining Software

The stratum proxy speaks plain Stratum V1, the same dialect `kawpowminer` uses, and is the only miner confirmed working against the network so far. Other KawPow-capable miners that auto-negotiate a different stratum dialect first (e.g. EthereumStratum/2.0) may fail to connect until they are configured to use plain Stratum V1 directly, if such an option exists in that miner.

- `kawpowminer`

6 Applications

6.1 Decentralized Gaming

Fair randomness for loot drops, matchmaking, and procedural generation.

6.2 Lotteries and Raffles

Draws whose result the operator cannot choose, subject to the bound in Section 4.2.

6.3 NFT Reveals

Random trait assignment without oracle dependency.

6.4 DAO Governance

Sortition committees selected via beacon output.

6.5 Scientific Simulations

Entropy source for Monte-Carlo methods requiring public verifiability.

7 Roadmap

7.1 Short-Term

- Formal verification of beacon aggregation.
- Additional RPC endpoints.
- Documentation expansion.

7.2 Medium-Term

- Light-client beacon verification.
- Optional VRF overlay for hybrid randomness.

8 Status and Limitations

Sharecoin is developed and maintained by a single contributor, with no formal team or external funding. The consensus changes described in this paper have not undergone external security audit; all review has been internal. The genesis block reuses Bitcoin’s original, provably unspendable coinbase output, so there is no premine and no admin-controlled token allocation. There are currently no exchange listings and no established market for SHC. The bias analysis in Section 4.2 was corrected on September 23, 2026: an earlier version overstated the effect of the window size.

9 Conclusion

Sharecoin demonstrates that GPU-based PoW can serve as a practical, publicly verifiable randomness beacon. By preserving Bitcoin Core compatibility and introducing a novel entropy primitive, Sharecoin provides a foundation for decentralized applications requiring publicly verifiable randomness with bounded bias, without trusted parties or external oracles.